

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	17.3 / 2.14452 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-07-07 12:00 UTC
Location	30.0395, -95.6739
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	4 / 375	39.12 to 100; mean 74.6099	100 at 2026-07-07 12:04Z; 4 min after event
asos	Air temperature (temperature)	degC	4 / 375	21.6667 to 35.5556; mean 28.571	24.4444 at 2026-07-07 12:04Z; 4 min after event
asos	Wind direction (wind_direction)	degree	4 / 508	0 to 360; mean 132.559	0 at 2026-07-07 11:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	4 / 540	0 to 15.4333; mean 2.3074	0 at 2026-07-07 11:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	5 / 65	5891.13 to 5923.27; mean 5907.53	5898.09 at 2026-07-07 12:00Z; at event time
noaa_gfs	Planetary boundary layer height (pbl_height)	m	5 / 65	93.8794 to 2239.24; mean 829.888	108.759 at 2026-07-07 12:00Z; at event time
noaa_gfs	Precipitable water (precipitable_water)	mm	5 / 65	38.7896 to 52.9161; mean 45.7018	45.7506 at 2026-07-07 12:00Z; at event time
noaa_gfs	Surface pressure (surface_pressure)	hPa	5 / 65	1004.8 to 1015.26; mean 1011.3	1010.52 at 2026-07-07 12:00Z; at event time
noaa_gfs	850-hPa temperature (t_850)	degC	5 / 65	17.4266 to 21.3086; mean 19.238	19.0207 at 2026-07-07 12:00Z; at event time
noaa_gfs	10-m eastward wind (u_10m)	m/s	5 / 65	-2.83889 to 3.62264; mean 0.5271	0.822056 at 2026-07-07 12:00Z; at event time
noaa_gfs	10-m northward wind (v_10m)	m/s	5 / 65	0.713226 to 4.9494; mean 2.2824	1.85059 at 2026-07-07 12:00Z; at event time
openaq	Ozone (ozone)	ppm	7 / 401	0 to 0.058; mean 0.0186	0 at 2026-07-07 12:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	1 / 62	14 to 60; mean 25.9516	25 at 2026-07-07 12:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	60 / 2229	0 to 62.1; mean 5.7514	10.7 at 2026-07-07 12:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	3 / 215	0 to 100; mean 30.7628	2 at 2026-07-07 11:59Z; at event time
openweather	Relative humidity (humidity)	percent	3 / 215	50 to 95; mean 74.6558	88 at 2026-07-07 11:59Z; at event time
openweather	Precipitation rate (precipitation)	mm/h	3 / 215	0 to 5.62; mean 0.0387	0 at 2026-07-07 11:59Z; at event time
openweather	Surface pressure (pressure)	hPa	3 / 215	1011 to 1019; mean 1015.67	1017 at 2026-07-07 11:59Z; at event time
openweather	Air temperature (temperature)	degC	3 / 215	23.78 to 36.05; mean 29.6576	26.7 at 2026-07-07 11:59Z; at event time
openweather	Wind direction (wind_direction)	degree	3 / 215	0 to 359; mean 192.228	226 at 2026-07-07 11:59Z; at event time
openweather	Wind speed (wind_speed)	m/s	3 / 215	0.45 to 8.39; mean 2.1146	1.78 at 2026-07-07 11:59Z; at event time
purpleair	PM2.5 (pm25)	ug/m3	44 / 3024	0 to 65.4; mean 7.6657	9.2 at 2026-07-07 12:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	6 / 6	0.0243723 to 0.0253077; mean 0.0249	0.0249091 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	8 / 8	0.000174097 to 0.000235341; mean 0.0002	0.000235341 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	4 / 4	3.63106e-05 to 4.30727e-05; mean 0	3.63106e-05 at 2026-07-07 18:57Z; 6 h 58 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	4 / 4	1 to 1; mean 1	1 at 2026-07-07 18:57Z; 6 h 58 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	8 / 8	-3.49304e-05 to 4.72775e-05; mean 0	9.05926e-06 at 2026-07-07 18:27Z; 6 h 27 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-07-07 18:27Z; 6 h 27 min after event
tceq	Carbon monoxide (co)	ppm	1 / 58	0.1 to 1; mean 0.381	0.3 at 2026-07-07 12:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	6 / 338	0 to 35.8; mean 6.297	17.3 at 2026-07-07 12:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	07-06 00Z 71.43, 06Z 89.15, 12Z 72.17, 18Z 54.54, 07-07 00Z 77.93, 06Z 90.82, 12Z 73.55, 18Z 61.28, 07-08 00Z 79.36, 06Z 93.05, 12Z 73.24, 18Z 61.86
asos	Air temperature (temperature)	07-06 00Z 28.9, 06Z 26.21, 12Z 29.91, 18Z 32.5, 07-07 00Z 28.29, 06Z 25.78, 12Z 29.34, 18Z 31.2, 07-08 00Z 25.51, 06Z 24.16, 12Z 29.93, 18Z 30.9
asos	Wind direction (wind_direction)	07-06 00Z 143.8, 06Z 119.8, 12Z 195.2, 18Z 156.4, 07-07 00Z 151.3, 06Z 91.11, 12Z 124.7, 18Z 107.6, 07-08 00Z 99.15, 06Z 48.18, 12Z 192.6, 18Z 172
asos	Wind speed (wind_speed)	07-06 00Z 3.196, 06Z 1.393, 12Z 2.326, 18Z 3.655, 07-07 00Z 2.369, 06Z 0.9646, 12Z 1.183, 18Z 2.752, 07-08 00Z 3.247, 06Z 0.6431, 12Z 2.359, 18Z 3.915
noaa_gfs	500-hPa geopotential height (gh_500)	07-06 00Z 5905, 06Z 5906, 12Z 5892, 18Z 5902, 07-07 00Z 5897, 06Z 5904, 12Z 5898, 18Z 5914, 07-08 00Z 5912, 06Z 5918, 12Z 5911, 18Z 5923, 07-09 00Z 5915
noaa_gfs	Planetary boundary layer height (pbl_height)	07-06 00Z 1082, 06Z 322, 12Z 242.2, 18Z 1803, 07-07 00Z 1159, 06Z 299.9, 12Z 192.3, 18Z 2015, 07-08 00Z 1029, 06Z 313.8, 12Z 196.8, 18Z 1851, 07-09 00Z 282.6
noaa_gfs	Precipitable water (precipitable_water)	07-06 00Z 41.76, 06Z 45.03, 12Z 42.57, 18Z 42.28, 07-07 00Z 46.39, 06Z 46.21, 12Z 46, 18Z 47.21, 07-08 00Z 48.26, 06Z 47.76, 12Z 46.73, 18Z 47.46, 07-09 00Z 46.47
noaa_gfs	Surface pressure (surface_pressure)	07-06 00Z 1007, 06Z 1010, 12Z 1011, 18Z 1011, 07-07 00Z 1009, 06Z 1012, 12Z 1013, 18Z 1013, 07-08 00Z 1011, 06Z 1013, 12Z 1014, 18Z 1014, 07-09 00Z 1011
noaa_gfs	850-hPa temperature (t_850)	07-06 00Z 20.58, 06Z 20.1, 12Z 19.77, 18Z 18.08, 07-07 00Z 19.54, 06Z 19.69, 12Z 18.82, 18Z 17.98, 07-08 00Z 18.96, 06Z 19.83, 12Z 18.99, 18Z 17.93, 07-09 00Z 19.81
noaa_gfs	10-m eastward wind (u_10m)	07-06 00Z 1.183, 06Z 1.236, 12Z 0.9409, 18Z 0.9998, 07-07 00Z -0.04144, 06Z 0.448, 12Z 0.8261, 18Z 0.1033, 07-08 00Z -1.109, 06Z 0.2756, 12Z 0.6952, 18Z 1.016, 07-09 00Z 0.2791
noaa_gfs	10-m northward wind (v_10m)	07-06 00Z 2.133, 06Z 2.725, 12Z 1.694, 18Z 1.411, 07-07 00Z 3.17, 06Z 2.306, 12Z 1.479, 18Z 1.583, 07-08 00Z 3.939, 06Z 2.816, 12Z 1.437, 18Z 1.89, 07-09 00Z 3.088
openaq	Ozone (ozone)	07-06 00Z 0.02714, 06Z 0.007919, 12Z 0.01424, 18Z 0.03617, 07-07 00Z 0.01446, 06Z 0.006405, 12Z 0.01285, 18Z 0.0339, 07-08 00Z 0.02865, 06Z 0.005765, 12Z 0.0045, 18Z 0.0335
openaq	PM10 (pm10)	07-06 00Z 22.33, 06Z 16.67, 12Z 21.83, 18Z 27.83, 07-07 00Z 19.5, 06Z 19.5, 12Z 29.17, 18Z 31.83, 07-08 00Z 26.5, 06Z 36, 12Z 51
openaq	PM2.5 (pm25)	07-06 00Z 5.182, 06Z 6.816, 12Z 5.996, 18Z 5.565, 07-07 00Z 4.813, 06Z 6.022, 12Z 6.55, 18Z 7.593, 07-08 00Z 4.453, 06Z 6.763, 12Z 5.689, 18Z 3.082, 07-09 00Z 0.9778
openweather	Cloud cover (cloud_cover)	07-06 00Z 94.21, 06Z 35.94, 12Z 32, 18Z 29.17, 07-07 00Z 21.47, 06Z 4.833, 12Z 59, 18Z 43.33, 07-08 00Z 17.5, 06Z 4.056, 12Z 8.765, 18Z 14
openweather	Relative humidity (humidity)	07-06 00Z 70.16, 06Z 87.18, 12Z 78.78, 18Z 57, 07-07 00Z 74.32, 06Z 87.56, 12Z 76.44, 18Z 57.72, 07-08 00Z 76.5, 06Z 90.94, 12Z 80.41, 18Z 59.29
openweather	Precipitation rate (precipitation)	07-06 00Z 0.01105, 06Z 0, 12Z 0, 18Z 0.008889, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.4011, 06Z 0, 12Z 0, 18Z 0.04235
openweather	Surface pressure (pressure)	07-06 00Z 1012, 06Z 1014, 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1016
openweather	Air temperature (temperature)	07-06 00Z 30.4, 06Z 26.53, 12Z 29.45, 18Z 34.69, 07-07 00Z 29.72, 06Z 26.56, 12Z 29.93, 18Z 34.09, 07-08 00Z 27.4, 06Z 24.19, 12Z 29.3, 18Z 33.6
openweather	Wind direction (wind_direction)	07-06 00Z 169.9, 06Z 236.4, 12Z 233.8, 18Z 167.1, 07-07 00Z 174.7, 06Z 152.6, 12Z 214.9, 18Z 182.7, 07-08 00Z 166.9, 06Z 187.7, 12Z 229.1, 18Z 197.8
openweather	Wind speed (wind_speed)	07-06 00Z 2.565, 06Z 1.843, 12Z 1.778, 18Z 2.133, 07-07 00Z 1.586, 06Z 1.549, 12Z 1.344, 18Z 2.423, 07-08 00Z 3.151, 06Z 2.287, 12Z 2.262, 18Z 2.469
purpleair	PM2.5 (pm25)	07-06 00Z 7.163, 06Z 9.232, 12Z 7.331, 18Z 6.188, 07-07 00Z 6.718, 06Z 7.749, 12Z 8.462, 18Z 9.575, 07-08 00Z 7.072, 06Z 9.603, 12Z 7.677, 18Z 5.221
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1

Source	Metric	Values
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	CO column (s5p_co_column)	07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437
sentinel5p	CO granules available (s5p_co_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1
tceq	Carbon monoxide (co)	07-06 00Z 0.3333, 06Z 0.3667, 12Z 0.3167, 18Z 0.2667, 07-07 00Z 0.4, 06Z 0.3833, 12Z 0.3, 18Z 0.4833, 07-08 00Z 0.325, 06Z 0.5, 12Z 0.4833
tceq	Nitrogen dioxide (no2)	07-06 00Z 3.184, 06Z 4.365, 12Z 5.289, 18Z 5.064, 07-07 00Z 5.117, 06Z 4.753, 12Z 5.555, 18Z 8.483, 07-08 00Z 8.09, 06Z 10.85, 12Z 7.785, 18Z 1.1

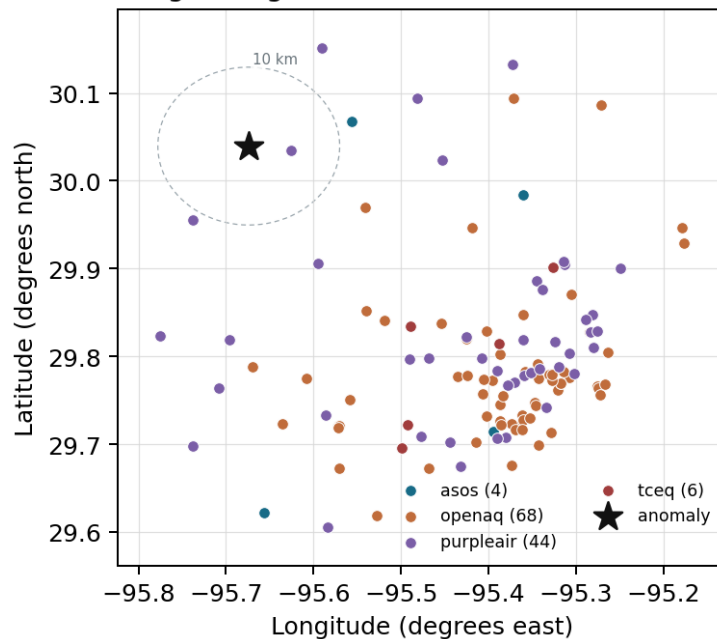
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

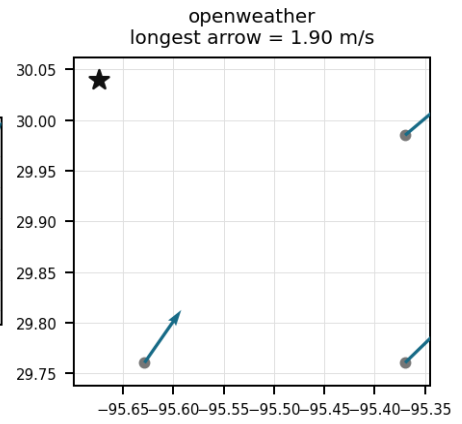
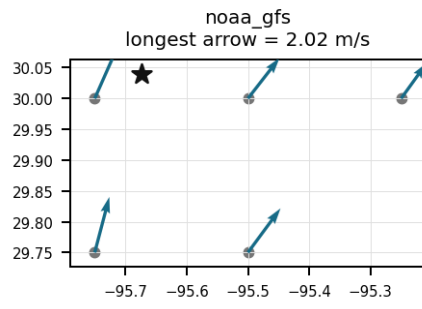
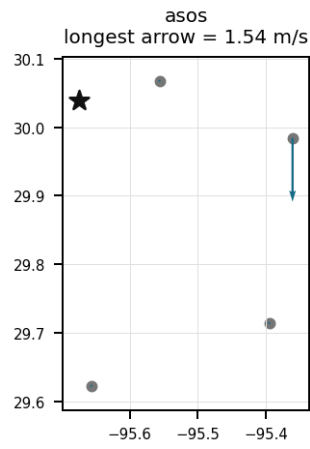
Source	Metric	Values
asos	Relative humidity (humidity)	DWH (11.767 km) 79.92, IAH (30.778 km) 71.97, MCJ (45.095 km) 72.16, SGR (46.424 km) 76.68
asos	Air temperature (temperature)	DWH (11.767 km) 26.98, IAH (30.778 km) 28.72, MCJ (45.095 km) 29.12, SGR (46.424 km) 28.81
asos	Wind direction (wind_direction)	DWH (11.767 km) 88.26, IAH (30.778 km) 152.4, MCJ (45.095 km) 134.5, SGR (46.424 km) 149.6
asos	Wind speed (wind_speed)	DWH (11.767 km) 1.292, IAH (30.778 km) 2.425, MCJ (45.095 km) 2.828, SGR (46.424 km) 2.55
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:30.00,-95.75 (8.54 km) 5907, gfs:30.00,-95.50 (17.314 km) 5908, gfs:29.75,-95.75 (33.018 km) 5908, gfs:29.75,-95.50 (36.299 km) 5908, gfs:30.00,-95.25 (41.053 km) 5907
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:30.00,-95.75 (8.54 km) 589.7, gfs:30.00,-95.50 (17.314 km) 981.8, gfs:29.75,-95.75 (33.018 km) 656, gfs:29.75,-95.50 (36.299 km) 922.9, gfs:30.00,-95.25 (41.053 km) 999
noaa_gfs	Precipitable water (precipitable_water)	gfs:30.00,-95.75 (8.54 km) 44.79, gfs:30.00,-95.50 (17.314 km) 46.01, gfs:29.75,-95.75 (33.018 km) 44.64, gfs:29.75,-95.50 (36.299 km) 45.51, gfs:30.00,-95.25 (41.053 km) 47.56
noaa_gfs	Surface pressure (surface_pressure)	gfs:30.00,-95.75 (8.54 km) 1009, gfs:30.00,-95.50 (17.314 km) 1011, gfs:29.75,-95.75 (33.018 km) 1011, gfs:29.75,-95.50 (36.299 km) 1013, gfs:30.00,-95.25 (41.053 km) 1012
noaa_gfs	850-hPa temperature (t_850)	gfs:30.00,-95.75 (8.54 km) 19.38, gfs:30.00,-95.50 (17.314 km) 19.21, gfs:29.75,-95.75 (33.018 km) 19.29, gfs:29.75,-95.50 (36.299 km) 19.19, gfs:30.00,-95.25 (41.053 km) 19.13
noaa_gfs	10-m eastward wind (u_10m)	gfs:30.00,-95.75 (8.54 km) 0.9249, gfs:30.00,-95.50 (17.314 km) 0.8442, gfs:29.75,-95.75 (33.018 km) 0.6196, gfs:29.75,-95.50 (36.299 km) 0.4757, gfs:30.00,-95.25 (41.053 km) -0.2289
noaa_gfs	10-m northward wind (v_10m)	gfs:30.00,-95.75 (8.54 km) 2.415, gfs:30.00,-95.50 (17.314 km) 2.051, gfs:29.75,-95.75 (33.018 km) 2.491, gfs:29.75,-95.50 (36.299 km) 2.423, gfs:30.00,-95.25 (41.053 km) 2.032

Source	Metric	Values
openaq	Ozone (ozone)	3723 (0.015 km) 0.02044, 319 (28.951 km) 0.01559, 287 (35.354 km) 0.01788, 310 (36.873 km) 0.02103, 1319333 (39.972 km) 0.02117, 311 (41.77 km) 0.01616, 2046 (44.308 km) 0.01781
openaq	PM2.5 (pm25)	14140749 (15.005 km) 7.004, 14140750 (24.564 km) 7.032, 14157977 (26.633 km) 6.958, 10762809 (26.733 km) 10.13, 14140753 (27.9 km) 6.505, 14103803 (29.758 km) 3.999, 14152860 (30.129 km) 2.008, 12337668 (30.849 km) 7.339, +52 more
openweather	Cloud cover (cloud_cover)	grid:north (29.906 km) 36.18, grid:west (31.345 km) 25.99, grid:center (42.695 km) 30.11
openweather	Relative humidity (humidity)	grid:north (29.906 km) 73.22, grid:west (31.345 km) 75.33, grid:center (42.695 km) 75.42
openweather	Precipitation rate (precipitation)	grid:north (29.906 km) 0.01306, grid:west (31.345 km) 0.02028, grid:center (42.695 km) 0.08324
openweather	Surface pressure (pressure)	grid:north (29.906 km) 1016, grid:west (31.345 km) 1016, grid:center (42.695 km) 1016
openweather	Air temperature (temperature)	grid:north (29.906 km) 29.52, grid:west (31.345 km) 29.57, grid:center (42.695 km) 29.89
openweather	Wind direction (wind_direction)	grid:north (29.906 km) 183.9, grid:west (31.345 km) 178.2, grid:center (42.695 km) 214.8
openweather	Wind speed (wind_speed)	grid:north (29.906 km) 2.275, grid:west (31.345 km) 2.041, grid:center (42.695 km) 2.026
purpleair	PM2.5 (pm25)	217323 (4.664 km) 6.251, 3033 (11.204 km) 4.678, 288282 (14.805 km) 7.586, 130763 (16.666 km) 2.604, 194469 (19.477 km) 10.57, 165203 (21.37 km) 5.381, 98139 (24.624 km) 7.575, 223797 (25.981 km) 5.672, +36 more
tceq	Nitrogen dioxide (no2)	482010029 (0.0 km) 2.396, 482010047 (28.955 km) 7.537, 482010024 (36.874 km) 6.089, 482011052 (37.243 km) 13.16, 482011066 (39.435 km) 5.461, 482010055 (41.774 km) 2.706

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

TCEQ station 482010029 recorded an NO2 concentration peak of 17.3 ppb at 12:00 UTC on July 7, 2026, yielding a z-score of 7.52 relative to the baseline expectation of 2.14 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 2 of 35

Increased vehicle emissions during morning rush hour at 12:00 UTC contributed to the localized spike in surface nitrogen dioxide concentrations observed by TCEQ monitors.

Mark exactly one:	☐ V	☐ I	☐ U
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Note (optional):

Claim 3 of 35

A localized primary NO_x plume is supported because the anomalous TCEQ monitor measured 17.3 ppb NO₂ at 2026-07-07T12:00:00+00:00 while the same monitor's mean was only 2.396 ppb and the 6-monitor TCEQ 12Z network mean was 5.555 ppb, indicating a site-specific enhancement rather than a uniform metropolitan increase.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 4 of 35

The 2026-07-07 12Z 6-hour mean NO₂ across the tceq monitors in the 50 km context window was 5.555 ppb, which is much lower than the 17.3 ppb reading at the anomalous monitor.

Mark exactly one:

☐ V

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Note (optional):

Claim 5 of 35

A shallow planetary boundary layer height of approximately 192 meters combined with light surface wind speeds trapped ground-level nitrogen dioxide emissions, causing the sharp concentration increase recorded by TCEQ.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 6 of 35

The observed NO₂ concentration of 17.3 ppb at 2026-07-07T12:00:00+00:00 represented a z-score elevation of 7.52 above the expected value of 2.14 ppb.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 7 of 35

NOAA GFS model data and ASOS meteorological observations near 12:00 UTC on July 7, 2026, documented a low planetary boundary layer height around 108 to 192 meters and calm surface winds averaging 1.18 m/s.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 8 of 35

The tceq monitor at coordinates 30.040, -95.674 recorded NO₂ of 17.3 ppb at 2026-07-07T12:00:00+00:00.

Mark exactly one:

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Note (optional):

Claim 9 of 35

There is no significant elevation in other pollutants such as CO, NO₂, SO₂, and O₃ at the time of the anomaly.

Mark exactly one:

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Note (optional):

Claim 10 of 35

The cause of the anomaly is likely due to a combination of factors, including industrial activities and vehicle emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 11 of 35

Local recirculation or drainage-scale transport is a likely contributing cause of the NO₂ spike because wind directions across nearby observations were variable while speeds were near calm, a setup that can steer a narrow urban or industrial plume onto one monitor without producing a similarly large regional enhancement.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 12 of 35

Stagnant surface air conditions and weak horizontal transport reduced pollutant dispersion, allowing local combustion emissions to concentrate near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 13 of 35

Meteorological conditions near 2026-07-07T12:00:00+00:00 favored near-surface accumulation of fresh NO_x because winds were very weak and the modeled planetary boundary layer height was very shallow at about 109 m.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 14 of 35

The air quality anomaly in Houston, Texas is caused by high levels of PM_{2.5}.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 15 of 35

A nearby fresh combustion source such as roadway traffic, diesel equipment, or an industrial combustion plume is a likely physical cause of the 17.3 ppb NO₂ spike because the anomalous monitor value was far above its own expected level and above the surrounding TCEQ network at the same time while co-located ozone was 0.0 ppm, a pattern consistent with a fresh local NO_x plume before photochemical dilution.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 16 of 35

A severe NO₂ concentration anomaly of 17.3 ppb was recorded at coordinates (30.040, -95.674) on 2026-07-07T12:00:00+00:00.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 35

The temperature inversion over the Houston area is causing a buildup of pollutants near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 35

The anomaly was characterized by high levels of PM_{2.5} and NO₂ in the air.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 35

A broad regional combustion or aerosol-rich episode is contradicted because co-located ozone in openaq was 0.0 ppm at 2026-07-07T12:00:00+00:00, regional PM2.5 remained modest at 6.55 ug/m3 in openaq 12Z means and 8.462 ug/m3 in purpleair 12Z means, and Sentinel-5P later the same day showed ordinary-scale CO and NO2 columns without a pronounced column enhancement over Houston.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 20 of 35

The air quality anomaly is related to particulate matter (PM2.5) with a value of 9.2 ug/m3.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 21 of 35

The temperature at the time of the anomaly was around 26.7°C.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 22 of 35

The combination of stagnant air and high humidity is trapping pollutants near the surface, exacerbating the air quality anomaly.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 23 of 35

The air quality in Houston, Texas was anomalous on July 7th, 2026.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 24 of 35

Weak dispersion is a likely contributing cause of the NO₂ spike because surface winds near the event were very light and the modeled planetary boundary layer was unusually shallow near the anomaly time, conditions that favor accumulation of near-surface primary NO₂ from local sources.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 35

Satellite observations from Sentinel-5P recorded modest column NO₂ levels around 3.87e-05 mol/m² during afternoon passes, confirming that the earlier anomaly was primarily a localized surface accumulation rather than a widespread regional column enhancement.

Mark exactly one:

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Note (optional):

Claim 26 of 35

Sentinel-5P satellite retrievals recorded low overall NO₂ column density values near 3.87e-5 mol/m² over Houston on July 7, 2026, indicating a localized near-surface accumulation rather than a widespread atmospheric plume.

Mark exactly one:

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Note (optional):

Claim 27 of 35

The air quality anomaly in Houston, Texas is not caused by high levels of NO₂.

Mark exactly one:

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Note (optional):

Claim 28 of 35

The NO2 anomaly at the tceq monitor at coordinates 30.040, -95.674 had an expected value of 2.1445205479452056 ppb and a z-score of 7.516590422226874, which is classified as severe.

Mark exactly one:

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Note (optional):

Claim 29 of 35

Ground-level air monitoring by TCEQ recorded an elevated NO2 concentration of 17.3 ppb at 12:00 UTC on July 7, 2026, representing a severe z-score anomaly of 7.52 relative to the expected baseline of 2.14 ppb.

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Note (optional):

Claim 30 of 35

Meteorological measurements from ASOS and GFS show near-stagnant surface winds and a shallow planetary boundary layer height near 108 to 192 meters around 12:00 UTC on July 7, 2026, which restricted vertical mixing of morning emissions.

Mark exactly one:

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Note (optional):

Claim 31 of 35

A nearby industrial facility or power plant is releasing excessive amounts of particulate matter and volatile organic compounds into the atmosphere.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 32 of 35

The TCEQ NO₂ monitor at 30.040, -95.674 recorded a severe local anomaly of 17.3 ppb at 2026-07-07T12:00:00+00:00 compared with an expected value of about 2.14 ppb, indicating an unusually large local NO₂ enhancement at that site.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 35

The air quality anomaly in Houston, Texas is not caused by high levels of SO₂.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 34 of 35

Regional co-pollutant and satellite evidence does not support a broad combustion or smoke episode because ozone at the nearest sensor was 0.0 ppm at the event time, PM2.5 remained only modestly elevated, TCEQ CO was 0.3 ppm, and the later Sentinel-5P NO2 and CO columns were not unusually high.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 35

Boundary-layer trapping is supported because the Houston-area morning environment at 2026-07-07T12:00:00+00:00 had very weak winds of about 1.18 m/s in asos 12Z means and 1.34 m/s in openweather 12Z means, alongside a very shallow GFS planetary boundary layer height near 108.8 m at the nearest grid cell and a 12Z mean near 192 m, which is consistent with poor dilution of fresh NOx emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
